Galatea: Supplementary Material

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Equation, proposition and section numbers refer to the main paper.

APPENDIX: PROOFS

Proof of Proposition 1: The objective of (4) is strictly concave: the entropy is strictly concave, and the penalty $-\tau\,\mathrm{KL}(\mathbf{P}^\top\mathbf{1}\|\mathbf{b})$ is concave because KL is convex in its first argument and $\mathbf{P} \mapsto \mathbf{P}^\top\mathbf{1}$ is linear. The feasible set $\{\mathbf{P} \geq 0,\ \mathbf{P}\mathbf{1} = \mathbf{a}\}$ is a product of scaled simplices, hence non-empty, convex and compact, and the objective is continuous on it (with $0\log 0 = 0$), so a maximiser exists; by strict concavity it is unique. Because $\partial(-P\log P)/\partial P \to +\infty$ as $P \to 0$, the maximiser has $P_{ij} > 0$, so only the equality constraints are active. With multipliers f_i for the row constraints (Lagrangian term $\sum_i f_i(\sum_j P_{ij} - a_i)$) and $\mathbf{q} = \mathbf{P}^\top\mathbf{1}$, stationarity gives $\bar{S}_{ij} - \varepsilon\log P_{ij} - \tau\log(q_j/b_j) + f_i = 0$, that is, $P_{ij} = \exp\big((\bar{S}_{ij} + f_i + g_j)/\varepsilon\big)$ with $g_j := \tau\log(b_j/q_j)$. Summing over i gives $q_j = e^{g_j/\varepsilon}\sum_i e^{(\bar{S}_{ij}+f_i)/\varepsilon}$. Substituting into the definition of g_j gives $g_j(1+\tau/\varepsilon) = \tau[\log b_j - \mathrm{LSE}_i((\bar{S}_{ij}+f_i)/\varepsilon)]$, which is (6) with $\lambda = \tau/(\tau+\varepsilon)$. The row constraint $\sum_j P_{ij} = a_i$ is (5). (i) Let Ψ be the composed map of Proposition 2, which is λ-Lipschitz. At the fixed point, $\|\mathbf{g}^\star\|_\infty \leq \|\Psi(\mathbf{g}^\star) - \Psi(\mathbf{0})\|_\infty + \|\Psi(\mathbf{0})\|_\infty \leq \lambda\|\mathbf{g}^\star\|_\infty + \|\Psi(\mathbf{0})\|_\infty$, and $\|\Psi(\mathbf{0})\|_\infty = O(\lambda)$ by (6). Hence $\|\mathbf{g}^\star\|_\infty \leq \|\Psi(\mathbf{0})\|_\infty/(1-\lambda) = O(\lambda)$, so $\mathbf{g}^\star \to \mathbf{0}$ as $\lambda \to 0$. Substituting $\mathbf{g} = \mathbf{0}$ into (5) gives $P_{ij} = a_i e^{\bar{S}_{ij}/\varepsilon}/\sum_k e^{\bar{S}_{ik}/\varepsilon}$. At $\tau = 0$ this holds exactly, because the problem decouples over rows into independent entropy-regularised linear programs on scaled simplices. (ii) As $\tau \to \infty$, $\tau\,\mathrm{KL}(\cdot\|\mathbf{b})$ Γ-converges to the indicator of $\{\mathbf{P}^\top\mathbf{1} = \mathbf{b}\}$. Since the feasible set is compact and the maximisers unique, $\mathbf{P}^\tau \to \mathbf{P}^{\mathrm{bal}}$ [1]. Setting $\lambda = 1$ in (5)–(6) gives the balanced Sinkhorn equations. □

Proof of Proposition 2: For any vector $\mathbf{x}$ and perturbation $\boldsymbol{\delta}$, $|\mathrm{LSE}(\mathbf{x}+\boldsymbol{\delta}) - \mathrm{LSE}(\mathbf{x})| \leq \|\boldsymbol{\delta}\|_\infty$, because $\nabla\mathrm{LSE}$ is a probability vector. Hence F is 1-Lipschitz in $\|\cdot\|_\infty$, G is λ-Lipschitz, and $\Psi = G \circ F$ is λ-Lipschitz. For $\lambda < 1$, Ψ is a contraction on $(\mathbb{R}^{m+1}, \|\cdot\|_\infty)$, so by Banach's theorem it has a unique fixed point $\mathbf{g}^\star$ (the potential of Proposition 1), and $\mathbf{g}^\star = G(\mathbf{f}^\star)$ with $\mathbf{f}^\star = F(\mathbf{g}^\star)$. Algorithm 1 computes $\mathbf{g}^{(l)} = G(\mathbf{f}^{(l-1)})$ and $\mathbf{f}^{(l)} = F(\mathbf{g}^{(l)})$ for $l = 1, \ldots, L$. Therefore $\|\mathbf{g}^{(l)} - \mathbf{g}^\star\|_\infty \leq \lambda\|\mathbf{f}^{(l-1)} - \mathbf{f}^\star\|_\infty$ and $\|\mathbf{f}^{(l)} - \mathbf{f}^\star\|_\infty \leq \|\mathbf{g}^{(l)} - \mathbf{g}^\star\|_\infty$; chaining these inequalities gives the bound. The rate does not depend on ε, n or m, because the Lipschitz constants do not. □

Proof of Proposition 3: After (5) with $a_i = 1$, $e^{f_i/\varepsilon} = 1/\sum_{j\leq m+1} e^{(\bar{S}_{ij}+g_j)/\varepsilon}$, so $P_{i,m+1} = e^{(z+g_{m+1})/\varepsilon}\big/\big(e^{(z+g_{m+1})/\varepsilon} + \sum_{j\leq m} e^{(s_{ij}+g_j)/\varepsilon}\big)$, which equals (7). The argument of the sigmoid is strictly decreasing in each s_{ij}, because $\partial\,\mathrm{LSE}/\partial x_j > 0$, and increasing in z. □

TABLE I
BACKBONE LEARNING RATE FOR JOINT FINE-TUNING, SELECTED ON THE VALIDATION SCENE (GALATEA HEAD, SEED 0, R@1 %).

Backbone lr	$1\cdot10^{-5}$	$3\cdot10^{-5}$	$1\cdot10^{-4}$	$3\cdot10^{-4}$	$1\cdot10^{-3}$
Val. R@1	68.4	70.4	76.6	68.9	18.4

Proof of Proposition 4: By (9), $\mathbf{V}_{j,:} = \sum_i P_{ij}\mathbf{h}_i$. Replacing P_{ij} by $c_j P_{ij}$ multiplies row j of $\mathbf{V}$ by $c_j > 0$, which intra-normalisation removes. The scene token does not depend on $\mathbf{P}$, so Φ is unchanged. For $P_{ij} = e^{(\bar{S}_{ij}+f_i+g_j)/\varepsilon}$, take $c_j = e^{-g_j/\varepsilon}$ to obtain (8), with $w_i = e^{f_i/\varepsilon}$ given by (5). For $\lambda = 0$, $\mathbf{g} = \mathbf{0}$ and $a_i = 1$. □

Proof of Proposition 6: $s(q,r) > s(q,r')$ is equivalent to $\beta\Delta > \delta$, which is (i). Since $0 \leq \phi \leq \phi_{\max}$, $\beta\Delta \leq \beta\phi_{\max}$, which gives (ii). Since ϕ is increasing, $s_\ell(q,r) \leq s_\ell(q,r')$ implies $\Delta \leq 0 \leq \delta$, which gives (iii). □

Proof of Proposition 5: For a similarity transform $\mathbf{u} \mapsto c\mathbf{R}\mathbf{u} + \mathbf{t}$, all pairwise distances scale by c and so does their mean μ, which leaves $\mathbf{C}^q$ unchanged. The appearance cost $\mathbf{M}$ does not depend on positions, and neither does the dustbin cost. This proves the claim for (10). The solver uses the positions only through $\mathbf{C}^q$ and $\mathbf{C}^c$ in (11), and its initial plan depends on $\mathbf{M}$ alone, so every iterate, the mutual-arg-max correspondences and their number are unchanged as well. A permutation of keypoints permutes rows and columns of $\mathbf{M}$, $\mathbf{C}$, $\mathbf{G}^{(\nu)}$ and $\mathbf{T}$ consistently; Sinkhorn scaling commutes with such permutations, so every iterate and the correspondences are permuted accordingly and their number is unchanged.□

REFERENCES

[1] L. Chizat, G. Peyré, B. Schmitzer, and F.-X. Vialard, "Scaling algorithms for unbalanced optimal transport problems," *Math. Comput.*, vol. 87, no. 314, pp. 2563–2609, 2018.

I. ADDITIONAL EXPERIMENTS

Backbone learning rate: For joint fine-tuning (Section VI-D) the backbone learning rate was selected on the validation scene only, with the Galatea head and seed 0 (Table I). The head learning rate, schedule, sampler and loss are those of the frozen-backbone runs; training lasts 1000 steps. Test results were not consulted. The largest rate diverges.

Cross-hardware reproduction: All frozen-backbone runs of the main paper were executed on a 2-core CPU. We re-ran them with identical code on NVIDIA Tesla P40 GPUs, after re-extracting the frozen features on the GPU (the zero-shot baselines reproduce exactly: 26.8, 28.1 and 35.8% R@1), and added two seeds for the four main heads (Table II). All means agree within one standard deviation, and every conclusion

TABLE II
CROSS-HARDWARE REPRODUCTION OF THE FROZEN-BACKBONE HEADS (TEST R@1, %, MEAN±STD). CPU: THE RUNS REPORTED IN THE PAPER (3 SEEDS). GPU: THE SAME CODE AND PROTOCOL RE-RUN ON NVIDIA TESLA P40 GPUS (5 SEEDS).

Head	CPU (3 seeds)	GPU (5 seeds)	GPU hard
GeM	31.5±0.4	32.3±1.1	22.1
NetVLAD	58.2±0.7	58.1±0.6	46.6
SALAD (λ=1)	59.4±0.7	60.0±0.6	48.8
Galatea, learned λ	59.8±0.8	59.7±1.0	47.8
$\lambda = 0$	59.9±0.8	60.1±0.9	48.1

of the paper holds. The single-seed ablations also reproduce: single-illumination training gives 47.8% (CPU: 47.3%).